Sri Lanka Institute of Information Technology

Faculty of Computing SLIIT

# Module: IT3010 – NDM
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# 2026 – Lab Report 05

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## Introduction

Simple Network Management Protocol (SNMP) is one of the most common network management protocols that can be used to manage and control network devices like servers, routers, switches, and printers. It helps network administrators gather information related to network devices, observe network performance, and detect network errors.

SNMP uses the manager/agent concept to operate. The network management system acts as the manager, and it communicates with the network device agents to obtain network management information stored in the Management Information Base (MIB) of the network device. This practical includes installing SNMP services, configuring SNMP on a Linux server, and using different SNMP commands to obtain network management information.

The main aim of this practical is to understand how SNMP can be used in a real network environment and how network administrators can utilize command-line interfaces to obtain network management information.

## Objectives

The goals of this practical are:

1. To install SNMP packages for use in a Linux environment.
2. To configure SNMP services on a server.
3. To understand the configuration files associated with SNMP.
4. To learn how to retrieve management information using SNMP commands.
5. To view the MIB tree using SNMP utilities.

## Tools and Environment

Here is the rewritten paragraph according to the rules provided:

Tools & Environment used for this practical

Linux Server

VMware Workstation

SNMP (Net-SNMP Package)

Terminal Command Line Interface

Network Services (DHCP, Named, SNMP)

# Procedure

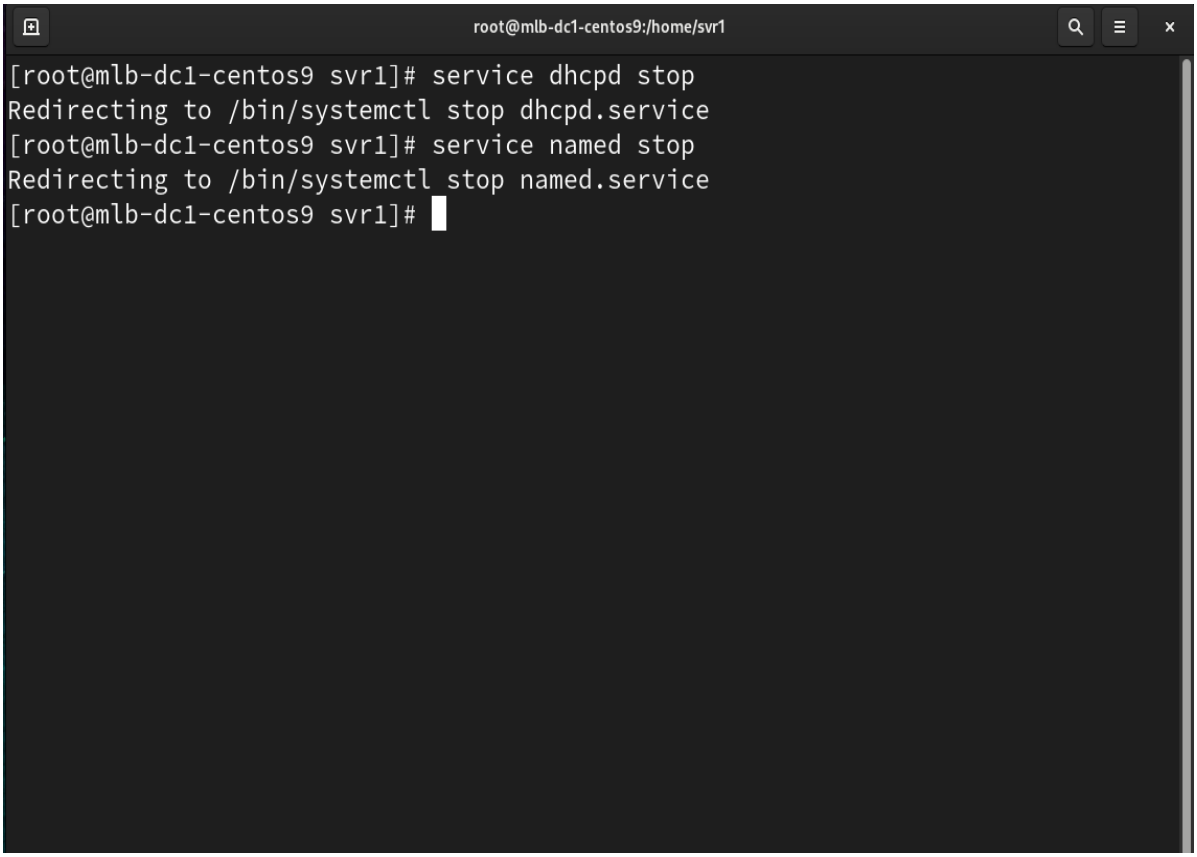
## Step 01 – Stopping Running Services

The initial steps involved terminating running services through the following command commands.

In order to terminate those running services, root privileges (superuser / admin) are mandatory.

Second, VMware network configuration was set up to VMnet8 (NAT) and the server configuration is to get an ip address via DHCP (dynamically assigned).

Finally, the network service was restarted and the assigned IP address was verified through DHCP and/or a static assignment.



**Check the DHCPD Status –** service dhcpd status

```
root@mlb-dc1-centos9:/home/svr1
○ dhcpd.service - DHCPv4 Server Daemon
     Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; enabled; preset: disabled)
     Active: inactive (dead) since Sat 2026-03-14 06:25:28 EDT; 1min 4s ago
   Duration: 4min 37.252s
       Docs: man:dhcpd(8)
             man:dhcpd.conf(5)
    Process: 1067 ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -grou>
   Main PID: 1067 (code=killed, signal=TERM)
     Status: "Dispatching packets..."
        CPU: 45ms

Mar 14 06:20:51 mlb-dc1-centos9.csa.lk dhcpd[1067]: Wrote 0 new dynamic host decls to lea>
Mar 14 06:20:51 mlb-dc1-centos9.csa.lk dhcpd[1067]: Wrote 1 leases to leases file.
Mar 14 06:20:51 mlb-dc1-centos9.csa.lk dhcpd[1067]: Listening on LPF/ens36/00:0c:29:2b:40>
Mar 14 06:20:51 mlb-dc1-centos9.csa.lk dhcpd[1067]: Sending on   LPF/ens36/00:0c:29:2b:40>
Mar 14 06:20:51 mlb-dc1-centos9.csa.lk dhcpd[1067]: Sending on   Socket/fallback/fallback>
Mar 14 06:20:51 mlb-dc1-centos9.csa.lk dhcpd[1067]: Server starting service.
Mar 14 06:20:51 mlb-dc1-centos9.csa.lk systemd[1]: Started DHCPv4 Server Daemon.
Mar 14 06:25:28 mlb-dc1-centos9.csa.lk systemd[1]: Stopping DHCPv4 Server Daemon...
Mar 14 06:25:28 mlb-dc1-centos9.csa.lk systemd[1]: dhcpd.service: Deactivated successfull>
Mar 14 06:25:28 mlb-dc1-centos9.csa.lk systemd[1]: Stopped DHCPv4 Server Daemon.
~
~
lines 1-21/21 (END)
```

**Check the Status-** service named status

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# service named status
Redirecting to /bin/systemctl status named.service
○ named.service - Berkeley Internet Name Domain (DNS)
     Loaded: loaded (/usr/lib/systemd/system/named.service; disabled; preset: disabled)
     Active: inactive (dead)
[root@mlb-dc1-centos9 svr1]#
```

**Check IP addresses using -** ifconfig

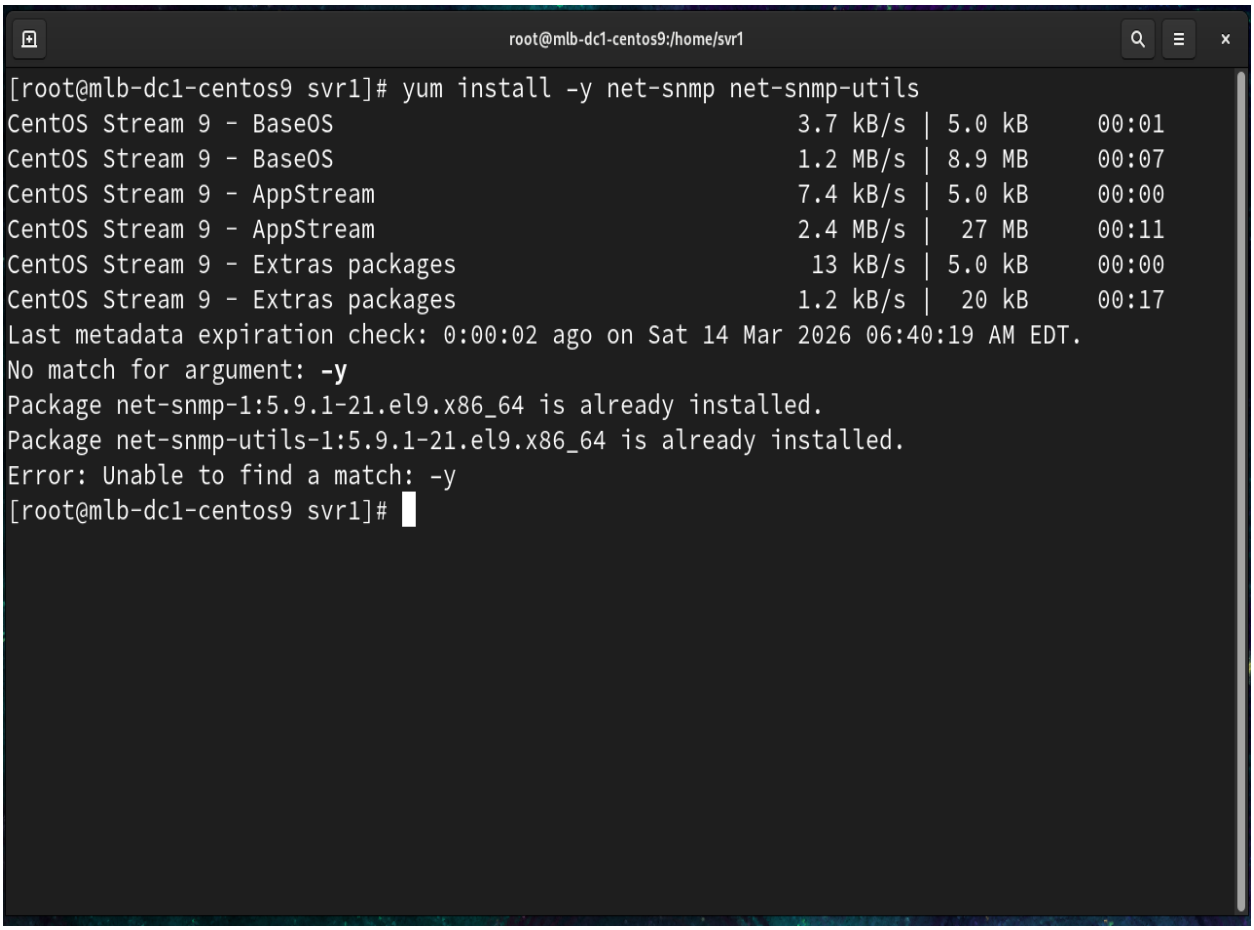
```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.234.128  netmask 255.255.255.0  broadcast 192.168.234.255
        inet6 fe80::20c:29ff:fe2b:4013  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:2b:40:13  txqueuelen 1000  (Ethernet)
        RX packets 883325  bytes 1283865808 (1.1 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 114905  bytes 6938194 (6.6 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ens36: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.1.2  netmask 255.255.255.0  broadcast 10.0.1.255
        inet6 fe80::1d81:51ca:cc03:340c  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:2b:40:1d  txqueuelen 1000  (Ethernet)
        RX packets 49  bytes 4547 (4.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 43  bytes 4733 (4.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 17  bytes 2099 (2.0 KiB)
```

**Step 02 – Install SNMP Packages:**

SNMP utilities were installed using the following command:

yum install –y net-snmp net-snmp-utils

This command installs the SNMP agent and the utilities required for network monitoring.



**Step 03 – Manual IP Configuration**

The installation of SNMP utilities can be accomplished with this command.

You will have installed the SNMP agent as well as the supporting components necessary for network monitoring.

The Network Adapter in your Virtual Machine will need to be changed to Host-Only Network, then saved when prompted.

Set up your physical server as follows:

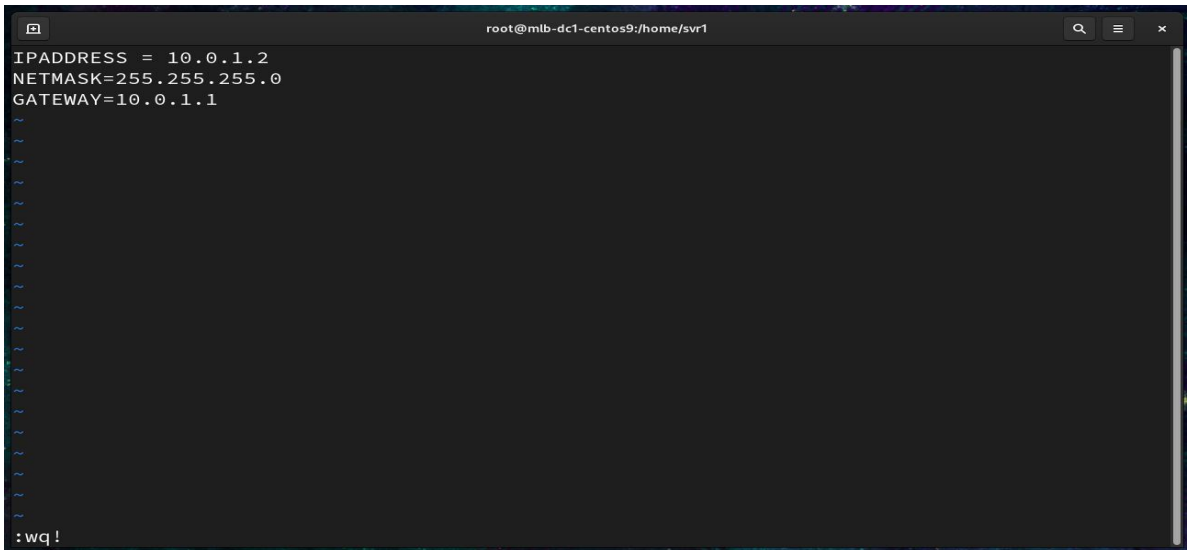
IP: 10.0.1.2

Netmask: 255.255.255.0

Gateway: 10.0.1.1

The final step in completing your configuration is to restart the network service after making these changes

vi /etc/sysconfig/network-scripts/ifcfg-ens36



Save and Quit – wq!

Restart Network

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# systemctl restart NetworkManager
[root@mlb-dc1-centos9 svr1]#
```

Check Status

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# systemctl restart NetworkManager
[root@mlb-dc1-centos9 svr1]# systemctl status NetworkManager
● NetworkManager.service - Network Manager
     Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled; preset: ena>
     Active: active (running) since Sat 2026-03-14 06:57:27 EDT; 54s ago
       Docs: man:NetworkManager(8)
   Main PID: 2750 (NetworkManager)
      Tasks: 3 (limit: 10717)
     Memory: 6.9M
        CPU: 408ms
     CGroup: /system.slice/NetworkManager.service
             └─2750 /usr/sbin/NetworkManager --no-daemon

Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9004] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9106] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9147] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9231] ma>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9348] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9391] ma>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9521] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9612] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9700] de>
Mar 14 06:57:30 mlb-dc1-centos9.csa.lk NetworkManager[2750]: <info>  [1773485850.9741] ma>
lines 1-21/21 (END)
```

## Step 04 – Configure SNMP:

vi /etc/snmp/snmpd.conf

Go to the down of the file and using “i” insert mode and add this code – Save and Quit : “wq!”

```
root@mlb-dc1-centos9:/home/svr1
# enterprises.ucdavis.255.5 = 42
# enterprises.ucdavis.255.6 = Gauge: 42
#
# % snmpget -v 1 localhost public .1.3.6.1.4.1.2021.255.5
# enterprises.ucdavis.255.5 = 42
#
# % snmpset -v 1 localhost public .1.3.6.1.4.1.2021.255.1 s "New string"
# enterprises.ucdavis.255.1 = "New string"
#

# For specific usage information, see the man/snmpd.conf.5 manual page
# as well as the local/passtest script used in the above example.

###############################################################################
# Further Information
#
#  See the snmpd.conf manual page, and the output of "snmpd -H".

rocommunity  public  10.0.1.2
rocommunity  public   127.0.0.1

syslocation  "HYD, UM DataCenter"
syscontact  admin@ndm.lk
"/etc/snmp/snmpd.conf" 468L, 18988B                                468,25        Bot
```

Start SNMP service using the following command : “service snmpd start” and Check the Status

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# service snmpd start
Redirecting to /bin/systemctl start snmpd.service
[root@mlb-dc1-centos9 svr1]# service snmpd status
Redirecting to /bin/systemctl status snmpd.service
● snmpd.service - Simple Network Management Protocol (SNMP) Daemon.
     Loaded: loaded (/usr/lib/systemd/system/snmpd.service; disabled; preset: disabled)
     Active: active (running) since Sat 2026-03-14 07:32:15 EDT; 5s ago
   Main PID: 3183 (snmpd)
      Tasks: 1 (limit: 10717)
     Memory: 9.9M
        CPU: 201ms
     CGroup: /system.slice/snmpd.service
             └─3183 /usr/sbin/snmpd -LS0-6d -f

Mar 14 07:32:14 mlb-dc1-centos9.csa.lk systemd[1]: Starting Simple Network Management Pro>
Mar 14 07:32:15 mlb-dc1-centos9.csa.lk snmpd[3183]: NET-SNMP version 5.9.1
Mar 14 07:32:15 mlb-dc1-centos9.csa.lk systemd[1]: Started Simple Network Management Prot>
lines 1-13/13 (END)
```

## SNMP Commands and Observations

## Step 05 – Explore the MIB Tree

I. **snmptranslate :** learning about the MIB tree. (translate MIB OID names between numeric and textual forms)

1. % snmptranslate .1.3.6.1.2.1.1.3.0

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptranslate .1.3.6.1.2.1.1.3.0
DISMAN-EVENT-MIB::sysUpTimeInstance
[root@mlb-dc1-centos9 svr1]#
```

2. % snmptranslate -On SNMPv2-MIB::system.sysUpTime.0

```
[root@mlb-dc1-centos9 svr1]# snmptranslate -On SNMPv2-MIB::system.sysUpTime.0
.1.3.6.1.2.1.1.3.0
[root@mlb-dc1-centos9 svr1]#
```

3. % snmptranslate .iso.3.6.1.private.enterprises.2021.2.1.prNames.0
   % snmptranslate -On .iso.3.6.1.private.enterprises.2021.2.1.prNames.0

```
[root@mlb-dc1-centos9 svr1]# snmptranslate .iso.3.6.1.private.enterprises.2021.2.1.prNames.0
UCD-SNMP-MIB::prNames.0
[root@mlb-dc1-centos9 svr1]# snmptranslate -On .iso.3.6.1.private.enterprises.2021.2.1.prNames.0
.1.3.6.1.4.1.2021.2.1.2.0
[root@mlb-dc1-centos9 svr1]#
```

4. % snmptranslate -Of .iso.3.6.1.private.enterprises.2021.2.1.prNames.0

```
[root@mlb-dc1-centos9 svr1]# snmptranslate -Of .iso.3.6.1.private.enterprises.2021.2.1.prNames.0
.iso.org.dod.internet.private.enterprises.ucdavis.prTable.prEntry.prNames.0
[root@mlb-dc1-centos9 svr1]#
```

5. % snmptranslate -TB 'vacm.*table'

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptranslate -TB 'vacm.*table'
SNMP-VIEW-BASED-ACM-MIB::vacmViewTreeFamilyTable
SNMP-VIEW-BASED-ACM-MIB::vacmAccessTable
SNMP-VIEW-BASED-ACM-MIB::vacmSecurityToGroupTable
SNMP-VIEW-BASED-ACM-MIB::vacmContextTable
NET-SNMP-VACM-MIB::nsVacmAccessTable
[root@mlb-dc1-centos9 svr1]#
```

6. % snmptranslate -On -Td -Ib 'sys.*ime'

```
[root@mlb-dc1-centos9 svr1]# snmptranslate -On -Td -Ib 'sys.*ime'
.1.3.6.1.2.1.1.9.1.4
sysORUpTime OBJECT-TYPE
  -- FROM       SNMPv2-MIB
  -- TEXTUAL CONVENTION TimeStamp
  SYNTAX        TimeTicks
  MAX-ACCESS    read-only
  STATUS        current
  DESCRIPTION   "The value of sysUpTime at the time this conceptual
            row was last instantiated."
::= { iso(1) org(3) dod(6) internet(1) mgmt(2) mib-2(1) system(1) sysORTable(9) sysOREntry(1) 4
}
[root@mlb-dc1-centos9 svr1]#
```

7. % snmptranslate -Tp -IR system

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptranslate -Tp -IR system
+--system(1)
   |
   +-- -R-- String    sysDescr(1)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -R-- ObjID     sysObjectID(2)
   +-- -R-- TimeTicks sysUpTime(3)
   |  |
   |  +--sysUpTimeInstance(0)
   |
   +-- -RW- String    sysContact(4)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -RW- String    sysName(5)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -RW- String    sysLocation(6)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -R-- INTEGER   sysServices(7)
   |        Range: 0..127
   +-- -R-- TimeTicks sysORLastChange(8)
   |        Textual Convention: TimeStamp
```

I. **snmpget :** retrieving data from a host.

1. % snmpget -v2c -c public 10.0.1.2 system.sysUpTime.0

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpget -v2c -c public 10.0.1.2 system.sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (150306) 0:25:03.06
[root@mlb-dc1-centos9 svr1]#
```

2.% snmpget -v 2c -c public 10.0.1.2 system.sysUpTime.0

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpget -v 2c -c public 10.0.1.2 system.sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (240004) 0:40:00.04
```

II. **snmpgetnext :** retrieving unknown indexed data.

1. % snmpgetnext -v 2c -c demopublic test.net-snmp.org system.sysUpTime.0

   % snmpgetnext -v 2c -c public 10.0.1.2 system.sysUpTime.0

```
[root@mlb-dc1-centos9 svr1]# snmpgetnext -v 2c -c public 10.0.1.2 system.sysUpTime.0
SNMPv2-MIB::sysContact.0 = STRING: admin@ndm.lk
[root@mlb-dc1-centos9 svr1]#
```

2. % snmpgetnext -v 2c -c demopublic test.net-snmp.org system.sysContact.0

   **% snmpgetnext -v 2c -c public 10.0.1.2 system.sysContact.0**

```
[root@mlb-dc1-centos9 svr1]# snmpgetnext -v 2c -c public 10.0.1.2 system.sysContact.0
SNMPv2-MIB::sysName.0 = STRING: mlb-dc1-centos9.csa.lk
[root@mlb-dc1-centos9 svr1]#
```

3. % snmpgetnext -v 2c -c demopublic test.net-snmp.org system.sysUpTime

   **% snmpgetnext -v 2c -c public 10.0.1.2 system.sysUpTime**

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpgetnext -v 2c -c public 10.0.1.2 system.sysUpTime
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (278294) 0:46:22.94
[root@mlb-dc1-centos9 svr1]#
```

4. % snmpgetnext -v 2c -c demopublic test.net-snmp.org .1.3.6
   **% snmpgetnext -v 2c -c public 10.0.1.2 .1.3.6**

```
[root@mlb-dc1-centos9 svr1]# snmpgetnext -v 2c -c public 10.0.1.2 .1.3.6
SNMPv2-MIB::sysDescr.0 = STRING: Linux mlb-dc1-centos9.csa.lk 5.14.0-565.el9.x86_64 #1 SMP PREEM
PT_DYNAMIC Fri Feb 7 18:47:11 UTC 2025 x86_64
[root@mlb-dc1-centos9 svr1]#
```

# Additional SNMP Commands

I. **snmpwalk** : Used to retrieve multiple SNMP objects. :
   "snmpwalk -v2c -c public 10.0.1.2 system"

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpwalk -v2c -c public 10.0.1.2 system
SNMPv2-MIB::sysDescr.0 = STRING: Linux mlb-dc1-centos9.csa.lk 5.14.0-565.el9.x86_64 #1 SMP PREEM
PT_DYNAMIC Fri Feb 7 18:47:11 UTC 2025 x86_64
SNMPv2-MIB::sysObjectID.0 = OID: NET-SNMP-MIB::netSnmpAgentOIDs.10
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (663307) 1:50:33.07
SNMPv2-MIB::sysContact.0 = STRING: admin@ndm.lk
SNMPv2-MIB::sysName.0 = STRING: mlb-dc1-centos9.csa.lk
SNMPv2-MIB::sysLocation.0 = STRING: "HYD, UM DataCenter"
SNMPv2-MIB::sysORLastChange.0 = Timeticks: (6) 0:00:00.06
SNMPv2-MIB::sysORID.1 = OID: SNMP-FRAMEWORK-MIB::snmpFrameworkMIBCompliance
SNMPv2-MIB::sysORID.2 = OID: SNMP-MPD-MIB::snmpMPDCompliance
SNMPv2-MIB::sysORID.3 = OID: SNMP-USER-BASED-SM-MIB::usmMIBCompliance
SNMPv2-MIB::sysORID.4 = OID: SNMPv2-MIB::snmpMIB
SNMPv2-MIB::sysORID.5 = OID: SNMP-VIEW-BASED-ACM-MIB::vacmBasicGroup
SNMPv2-MIB::sysORID.6 = OID: TCP-MIB::tcpMIB
SNMPv2-MIB::sysORID.7 = OID: UDP-MIB::udpMIB
SNMPv2-MIB::sysORID.8 = OID: IP-MIB::ip
SNMPv2-MIB::sysORID.9 = OID: SNMP-NOTIFICATION-MIB::snmpNotifyFullCompliance
SNMPv2-MIB::sysORID.10 = OID: NOTIFICATION-LOG-MIB::notificationLogMIB
SNMPv2-MIB::sysORDescr.1 = STRING: The SNMP Management Architecture MIB.
SNMPv2-MIB::sysORDescr.2 = STRING: The MIB for Message Processing and Dispatching.
SNMPv2-MIB::sysORDescr.3 = STRING: The management information definitions for the SNMP User-base
d Security Model.
SNMPv2-MIB::sysORDescr.4 = STRING: The MIB module for SNMPv2 entities
```

II. **snmptable** : Displays SNMP table data. : “snmptable -v2c -c public 10.0.1.2 ifTable”

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptable -v2c -c public 10.0.1.2 ifTable
IF-MIB::ifTable: No entries
[root@mlb-dc1-centos9 svr1]#
```

III. **snmpset** : Used to modify SNMP variables. –
“snmpset -v2c -c private 127.0.0.1 system. sysName.0 s "MyServer"”
“snmpget -v2c -c public 10.0.1.2 system.sysName.0S”

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmpset -v2c -c private 127.0.0.1 system.sysName.0 s "MyServer"
SNMPv2-MIB::sysName.0 = STRING: MyServer
[root@mlb-dc1-centos9 svr1]# snmpget -v2c -c public 10.0.1.2 system.sysName.0
SNMPv2-MIB::sysName.0 = STRING: MyServer
[root@mlb-dc1-centos9 svr1]#
```

IV. **snmptrap** : Used to send SNMP alerts. –
“snmptrap -v2c -c public 10.0.1.3 " SNMPv2-MIB::coldStart”

```
root@mlb-dc1-centos9:/home/svr1
[root@mlb-dc1-centos9 svr1]# snmptrap -v2c -c public 10.0.1.3 '' SNMPv2-MIB::coldStart
[root@mlb-dc1-centos9 svr1]#
```

## Results

SNMP services were successfully set up and installed on a Linux server in this practical. Management information was retrieved and analyzed from the MIB tree using a variety of SNMP commands, including snmptranslate, snmpget, and snmpgetnext. The instructions effectively illustrated how SNMP tools may be used by network managers to monitor device information.

## Conclusion

The objective of this practical was to obtain a complete understanding of SNMP and its application in network management. Through the installation and configuration of SNMP on a Linux server, it was possible to investigate the functionality of various SNMP utilities that are used for monitoring devices in a network. This practical provided an understanding of the application of SNMP in obtaining system information, such as uptime, and configuration information from devices using the MIB structure. This exercise enabled the development of skills that are necessary for network management.